PV102 Day 2: Site and Solar Resource Assessment

Students convert vague site assumptions into observable site conditions: solar access, shade, orientation, tilt, space, weather exposure, access, roof or ground constraints, and maintenance realities.

PV102 | CTS Day 18 | Instructor Run Sheet
Measure the site before you size the system. | Tuesday, August 4, 2026

Core rule

Measure the site before you size the system.

Learning targets

- Identify site factors that affect PV production and installation practicality.
- Use orientation, tilt, shade, and access observations to support a design decision.
- Separate measured observations from assumptions and preferences.
- Create a site-note handoff that Day 3 can use for load and energy work.

Readiness and materials

Before students arrive	Materials
Open slide deck and test display. Print student packets and one instructor key. Prepare three team stations. Review safety boundary before any equipment is handled.	Compass or phone compass app Tape measures Clipboards Site sketch sheets Markers Scenario cards Optional shade objects: cardboard/tree props/ladder silhouette Projector and slide deck

Five-hour run of show

Time	Block	Instructor move
10:00-10:15	Opening and continuity	Connect Day 1 to today's rule. Name what students already know and what new judgment is being added.
10:15-10:40	Morning quiz and review	Students work independently for 12-15 minutes. Review by asking for evidence, not just letters.
10:40-11:25	Concept lesson	Use slides 1-14. Pause after each technical rule and ask what decision it changes.
11:25-12:15	Guided practice	Model one example on the board. Students complete a quick pair-check before lunch.
12:15-12:45	Lunch/reset	Reset lab stations. Keep energized work out of scope unless explicitly authorized by instructor and equipment procedure.

12:45-1:00	Lab brief	Assign teams and roles. Review acceptance criteria and stop-work triggers.
1:00-2:15	Site Walk and Solar Access Handoff	Teams inspect three candidate locations and decide which one deserves further design work.
2:15-2:40	Handoffs and peer verification	Each team gives a concise technical handoff. Peers check evidence, boundary, and unknowns.
2:40-3:00	Exit ticket and reset	Collect packets. Preview next step: PV102 Day 3 uses the site handoff to build a load profile and first-pass energy budget.

Instructor talk track

Opening script: "PV102 is not about throwing parts at a problem until reality files a complaint. Today we use site and solar resource assessment to make the next design decision safer and more defensible. The rule is simple: Measure the site before you size the system."

Teaching notes

Move	Look-for
Ask	What decision does this evidence change?
Listen for	Students separating facts, assumptions, and unknowns.
Correct gently	Component-first answers. Redirect to job, evidence, boundary, then component.
Require	A handoff that another person could act on without reading minds. Humanity keeps trying telepathy. It remains a poor project-management system.

Lab instructor procedure

- Assign one candidate location per team.
- Sketch the site from above and mark north, shade sources, access path, approximate array area, and likely wire route.
- Record observations in three columns: measured, observed, unknown.
- Name the best reason to keep or reject the site for first-pass design.
- Prepare a two-minute handoff for Day 3.

Answer snapshot and contingency plan

If equipment or printing is unavailable, run the lab with paper cards, chart paper, and oral handoffs. The learning target is decision quality, not hardware theater.

Safety and scope boundaries

No live electrical work is authorized by this lesson unless the instructor independently sets up a verified training system, follows manufacturer instructions, and establishes the required isolation, PPE, and supervision. Treat PV source circuits and batteries as energy sources, not props.

Day handoff

PV102 Day 3 uses the site handoff to build a load profile and first-pass energy budget.

Instructional sources

- U.S. Department of Energy, Solar Energy: https://www.energy.gov/topics/solar-energy
- U.S. Department of Energy, Solar Photovoltaic Technology Basics: https://www.energy.gov/eere/solar/solar-photovoltaic-technology-basics
- U.S. Department of Energy, Solar Photovoltaic System Design Basics: https://www.energy.gov/eere/solar/solar-photovoltaic-system-design-basics
- U.S. Department of Energy, Solar Integration: Solar Energy and Storage Basics: https://www.energy.gov/eere/solar/solar-integration-solar-energy-and-storage-basics
- U.S. Department of Energy, Solar Integration: Inverters and Grid Services Basics: https://www.energy.gov/eere/solar/solar-integration-inverters-and-grid-services-basics
- NREL PVWatts Calculator: https://pvwatts.nrel.gov/
- OSHA, Green Job Hazards - Solar Energy: Electrical: https://www.osha.gov/green-jobs/solar/electrical
- OSHA, Green Job Hazards - Solar Energy: Lockout/Tagout: https://www.osha.gov/green-jobs/solar/lockout-tagout
- OSHA, Green Job Hazards - Solar Energy: Personal Protective Equipment: https://www.osha.gov/green-jobs/solar/personal-protective-equipment
- Sean White, Solar Photovoltaic Basics, 2nd ed., Routledge, 2018, ISBN 9781138102866.

PV102 Day 2 Morning Quiz: Answer Key

10 questions | 15 minutes | Independent work

PV102 | CTS Day 18 | Morning Quiz - Answer Key
Measure the site before you size the system. | Tuesday, August 4, 2026

Directions: Answer from your current knowledge. For written responses, use complete technical statements. If unsure, state the safe next action rather than guessing.

1. Which site condition most directly reduces solar access?

- A. Good labeling
- B. Shade on the array area
- C. A clean classroom
- D. A larger clipboard

Answer: B

Rationale/look-for: Shade reduces irradiance reaching the module surface and can also create mismatch effects.

2. Azimuth describes:

- A. Direction/orientation
- B. Battery chemistry
- C. Wire color
- D. Breaker rating

Answer: A

Rationale/look-for: Azimuth is the compass direction a surface or array faces.

3. Tilt describes:

- A. Array angle relative to horizontal
- B. Fuse speed only
- C. Type of current
- D. Wire insulation color

Answer: A

Rationale/look-for: Tilt is the mounting angle of the module surface.

4. A site note should separate observations from:

- A. Measurements
- B. Evidence
- C. Assumptions
- D. Safety boundaries

Answer: C

Rationale/look-for: Assumptions must be labeled so later design decisions do not pretend guesses are facts.

5. The safest first action before a roof assessment is to:

- A. Climb quickly
- B. Discuss access, fall hazards, and boundaries
- C. Carry modules first
- D. Energize the inverter

Answer: B

Rationale/look-for: Access and fall hazards must be controlled before roof work.

6. Name one factor besides shade that affects PV placement.

_____

_____

Answer: Instructor judgment

Rationale/look-for: Accept orientation, tilt, available area, structure, roof condition, wiring route, access, weather exposure, or maintenance space.

7. A team records "probably enough sun." What should they do next?

_____

_____

Answer: Convert to evidence

Rationale/look-for: They should make a specific observation or record it as an unknown to verify.

8. Why can a sunny site still be a poor installation site?

_____

_____

Answer: Constraints

Rationale/look-for: Space, access, structure, wiring route, safety, maintenance, or client need may still make it unsuitable.

9. What is one unknown that should be carried into Day 3?

_____

_____

Answer: Any relevant unknown

Rationale/look-for: Examples: actual load size, duty cycle, roof structure, utility status, shading hours, or available mounting area.

10. Write the Day 2 rule in your own words.

_____

_____

Answer: Measure before sizing

Rationale/look-for: Look for the idea that site evidence comes before design confidence.

PV102 Day 2 Whiteboard Outline

PV102 | CTS Day 18 | Whiteboard Outline
Measure the site before you size the system. | Tuesday, August 4, 2026

Board 1: Job	Board 2: Evidence	Board 3: Handoff
Topic: Site and Solar Resource Assessment Rule: Measure the site before you size the system. Question: What job are we solving?	Measured / Observed / Unknown Constraint that matters most Safety boundary before work	Decision Evidence Risk Unknowns Next action

Scenario anchors

- South-facing roof plane with one morning tree shadow and limited attic access.
- Open ground rack area with excellent sun but long wire run and mower traffic.
- West-facing shed roof near the load with easy service access and afternoon sun.

Final board sentence

Measure the site before you size the system.

PV102 Day 2 Student Packet

Site and Solar Resource Assessment

Today's rule

Measure the site before you size the system.

Learning targets

- Identify site factors that affect PV production and installation practicality.
- Use orientation, tilt, shade, and access observations to support a design decision.
- Separate measured observations from assumptions and preferences.
- Create a site-note handoff that Day 3 can use for load and energy work.

Morning quiz

1. Which site condition most directly reduces solar access?

- A. Good labeling
- B. Shade on the array area
- C. A clean classroom
- D. A larger clipboard

2. Azimuth describes:

- A. Direction/orientation
- B. Battery chemistry
- C. Wire color
- D. Breaker rating

3. Tilt describes:

- A. Array angle relative to horizontal
- B. Fuse speed only
- C. Type of current
- D. Wire insulation color

4. A site note should separate observations from:

- A. Measurements
- B. Evidence
- C. Assumptions
- D. Safety boundaries

5. The safest first action before a roof assessment is to:

- A. Climb quickly
- B. Discuss access, fall hazards, and boundaries
- C. Carry modules first
- D. Energize the inverter

6. Name one factor besides shade that affects PV placement.

_____

_____

7. A team records "probably enough sun." What should they do next?

_____

_____

8. Why can a sunny site still be a poor installation site?

_____

_____

9. What is one unknown that should be carried into Day 3?

_____

_____

10. Write the Day 2 rule in your own words.

_____

_____

Reference: decision order

Step	Action	Proof question
1	Define the job	What must the system accomplish?
2	Collect evidence	What is measured, observed, or still unknown?
3	Choose boundary	What must be protected, isolated, sized, or verified?
4	Defend the choice	What evidence supports the decision?
5	Handoff	What should the next person know?

Site Walk and Solar Access Handoff

Teams inspect three candidate locations and decide which one deserves further design work.

Scenario options

- South-facing roof plane with one morning tree shadow and limited attic access.
- Open ground rack area with excellent sun but long wire run and mower traffic.
- West-facing shed roof near the load with easy service access and afternoon sun.

Team procedure

- Assign one candidate location per team.
- Sketch the site from above and mark north, shade sources, access path, approximate array area, and likely wire route.
- Record observations in three columns: measured, observed, unknown.
- Name the best reason to keep or reject the site for first-pass design.
- Prepare a two-minute handoff for Day 3.

Team worksheet

Prompt	Team response
System/job statement	
Evidence used	
Decision made	
Safety boundary	
Top unknowns	
Next action	

Exit ticket

Which site condition would most likely force a redesign, and what evidence supports that answer?

_____

_____

PV102 Day 2 Afternoon Lab

Site Walk and Solar Access Handoff

PV102 | CTS Day 18 | Afternoon Lab
Measure the site before you size the system. | Tuesday, August 4, 2026

Mission

Teams inspect three candidate locations and decide which one deserves further design work.

Safety boundary

This lab is training work. Do not energize or connect equipment outside the instructor-approved procedure. When in doubt, stop, isolate, verify, and ask.

Scenario options

- South-facing roof plane with one morning tree shadow and limited attic access.
- Open ground rack area with excellent sun but long wire run and mower traffic.
- West-facing shed roof near the load with easy service access and afternoon sun.

Procedure

- Assign one candidate location per team.
- Sketch the site from above and mark north, shade sources, access path, approximate array area, and likely wire route.
- Record observations in three columns: measured, observed, unknown.
- Name the best reason to keep or reject the site for first-pass design.
- Prepare a two-minute handoff for Day 3.

Acceptance criteria

- Match the real job.
- Show the energy or work path.
- Name the dominant constraint.
- State the safety boundary.
- Record unknowns instead of guessing.
- Give a handoff another crew can use.

Team handoff form

Item	Response
Team members	
Scenario	
Primary decision	

Evidence	
Safety boundary	
Unknowns	
Next safest action	

PV102 Day 2 Classroom Post

Copy-ready announcement and assignment text

PV102 | CTS Day 18 | Google Classroom Post
Measure the site before you size the system. | Tuesday, August 4, 2026

Announcement

Today in PV102 we are working on site and solar resource assessment. The core rule is: Measure the site before you size the system.

Assignment title

PV102 Day 2 - Site and Solar Resource Assessment

Student directions

- Complete the morning readiness quiz independently.
- Use the reference section during the afternoon lab.
- Work with your team to complete the scenario handoff.
- Complete the exit ticket before class ends.

Turn in

- Completed student packet
- Team lab handoff
- Exit ticket

Success criteria

Match the need. Show the path. Name the constraint. State the safety boundary. Record unknowns.